MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRIODE, POWER

TYPE 7835

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: UHF triode, pulsed amplifier, transmitting

PHYSICAL CHARACTERISTICS:

Mounting position:	Tube axis vertical, filament terminal up or down.
Cathode:	Multistrand, thoriated-tungsten filament.
Dimensional outline:	See figures 1, 2, 3, 4 and 5.
Weight:	190 pounds nominal.

ABSOLUTE RATINGS:
(Anode pulsed class B amplifier service)

Parameter:	A If	eb	Ec1	ib	ik	Ib	Ik	Pi	Pp
Unit:	Adc	kv	Vdc	a	a	Adc	Adc	kW	kW
Note:	2	3	-	-	-	-	-	-	-
Maximum:	7000	45	-250	300	500	3	5	135	85
Minimum:	-	-	-	-	-	-	-	-	-
Test Conditions:	6600	-	-	-	-	-	-	-	-

ABSOLUTE RATINGS:
(Anode pulsed class B amplifier service)

Parameter:	tk	tp	F_1	Cooling	Fault Protection
Unit:	sec	μs	MHz	-	-
Note:	2	4	5	6	1
Maximum:	-	25	300	-	-
Minimum:	30	-	-	-	-
Test Conditions:	-	-	-	6	-

GENERAL:

Qualification:	See note 7.
Holding period:	t = 96 hours

(A) Complete Revision

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PAR. NO.	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 1							
	Grid cathode voltage standoff	Ecl = 1000 Vdc No other voltages	0.65	II	----	Note 8	----	----
1256	Electrode current (anode)	Eb = -45 Vdc Ecl/Icl = 50 Adc t = 5 min See notes 9 and 15	0.65	II	Ib	----	-25	μAdc
1261	Electrode voltage (filament)	See note 10	0 65	II	Ef	3.1	4.2	Vdc
1155	Anode water pressure differential	No voltages See note 11	0.65	II	PPD	----	45	psi
1155	Upper grid water pressure differential	No voltages See note 11	0.65	II	UGPD	----	25	psi
1155	Lower grid water pressure differential	No voltages See note 11	0.65	II	LGPD	----	25	psi
1155	Grid-cathode water pressure differential	No voltages See note 11	0.65	II	GKPD	----	35	psi
----	Anode voltage standoff	Eb = 35 KVdc Ecl/Ib = .01 Adc	0.65	II	----	Note 12	35	kV
----	Power output	If = 7000 Adc (max) eb = 35 ±1 Kv F = 400 ±50 MHz t = 1 hour (min) Peak driver power= 300 kw (max) Ib = 400a (max) See note 13	0.65	II	Po	5	----	Mw
	Quality conformance inspection, part 2							
----	Input resonant frequency	See note 14	----	----	Fi	90	120	MHz
----	Output resonant frequency	See note 14	6.5	S3	Fo	240	280	MHz
1105	Permanance of marking	-----	----	----	----	----	----	----
	Quality conformance inspection, part 3							
----	Service life guarantee	Within absolute ratings	----	----	t	1000	---	hrs
---	Service life end points	Power output	----	----	Po	4	---	Mw

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NOTES:

1. Fault Protection

A high-speed electronic protective device shall be used to prevent energy from the anode power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the anode pulse source supplying the tube during the interpulse interval following the occurrence of a fault in any single pulse. Thus, all pulses subsequent to a "fault" pulse shall not be applied until the device is reset. Minimum reset time shall be one second. (A fault is a condition in which the peak pulse anode current rises 20 percent or more above the normal peak anode current level.) In addition, the grid drive line shall be provided with Voltage Standing Wave Ratio (VSWR) protection to remove drive power and anode voltage within 10 milliseconds in the event of abnormal changes in input VSWR during operation.

2. Filament Current

Time (tk) begins after filament voltage is raised to operating value. To avoid undue thermal stresses in the filament, it is recommended that the filament voltage be raised to operating value in not less than 30 seconds. Filament surge current shall never exceed 2000 amperes, even momentarily.

3. Anode Voltage

Pressurization of the output and input cavities may be required to prevent corna or external flashover at the ceramic bushings.

4. Pulse Duration

On time (tp) is defined as the sum of the durations of all individual pulses which occur during any 2500 microsecond interval.

5. Frequency

Operation above 300 MHz is possible at derated conditions; however, the bureau or agency concerned shall be consulted relating to such applications.

6. Cooling
Ratings

Water cooling of the grid-cathode coolant course, each grid-terminal coolant, and the anode is required. The water flow shall start before application of any voltages and shall continue for several minutes after removal of all voltages. Interlocking of the water flow for each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate water flow.

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NOTES: (Cont'd)

It is essential that the resistivity of the grid-cathode water coolant shall be no less than 5 megohm-cm. The resistivity of the anode water coolant shall be no less than 1 megohm-cm. Water flow requirements are as follows:

	Min. gpm	Typical gpm	Max. Pressure Differential (psi) for Typical Flow (See note below)	Max. Inlet Pressure (psig)
Anode Inlet (For Pp=85 kW)	90	100	30	90
Grid-Cathode Coolant Course	30	35	30	90
Upper Grid Coolant Course	2	3	25	90
Lower Grid Coolant Course	2	3	25	90

Note: Measured directly across the cooled element.

Maximum or minimum temperatures shall be as follows:

a. Water (at any outlet): 70°C max.
b. Ceramic-bushing: 150°C max.
c. Metal-surface: 150°C max.
d. Storage Temperature: -65°C min.

(Water cooled elements shall be free of water before storage or shipment to prevent damage from freezing.)

Test Conditions

The water flow to the anode coolant connection shall be 30 gpm minimum. The water flow to each grid coolant course shall be 2 gpm minimum. A minimum water flow of 30 gpm shall be applied to the grid-cathode coolant course. Water flow shall start before the application of any voltages and continue for at least 10 seconds after removal of all voltages. The outlet water temperature shall not exceed 70°C.

7. Qualification

All reference to qualification test in MIL-E-1 shall be deleted and replaced by preproduction tests. All tests specified herein shall be conducted on a preproduction sample which shall be representative of the production tubes. The first article test program is designed to demonstrate the manufacturer's capability to meet the performance specification on the tube. Therefore, the first article requirement is applicable only on the first procurement from any supplier. On all subsequent procurements, the first article requirement will not apply.

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NOTES: (Cont'd)

8. Grid Cathode Voltage Standoff

Specified voltage shall be applied between grid and cathode. The tube shall be acceptable if there is no indication of internal short during a one-minute interval. The grid cathode coolant course shall be dry before the initiation of this test.

9. Plate Current

After the specified time, record the current from the anode supply as I1, reduce EC1 to zero and record the water column leakage current as I2, Anode Current, -Ib = I1 -I2.

10. Filament Voltage

Filament Voltage shall be measured between the RF cathode terminals.

11. Water Pressure Differential

Anode Water Pressure Differential:
The water flow to the anode coolant connection shall be 160 gpm.

Upper and Lower Grid Water Differential:
The water flow to the grid coolant course shall be 3 gpm.

Grid-Cathode Water Pressure Differential:
The water flow to the grid-cathode collant course shall be 40 gpm.

12. Plate Voltage Standoff

The tube shall be acceptable if the anode current (Ib) does not exceed 0.01 ampere during a one-minute interval.

13. Power Output

Water flow shall not be less than 100 gpm (anode) 35 gpm (grid-cathode) and 3 gpm each (upper and lower grid). Test is an anode pulsed, grounded grid operation performed at a 25 μsec minimum pulse length and a duty factor of 0.015 minimum. (Duty factor is the product of pulse duration and repetition rate.) Bias voltage shall be developed across a suitable cathode resistor and shall be adjusted for class B operation. The tube shall run for one hour at the specified conditions without a tripout. (A tripout is defined as four consecutive missing pulses attributable to the tube.) A maximum of 100 missing pulses is allowable.

The grid drive coupling and output loading shall be adjusted to obtain the specified operating conditions. Power output shall be measured across a water cooled resistive load. Power output shall be determined as follows:

$$Po(\text{kilowatts}) = \frac{0.264\ n\ (to - ti)}{\text{duty factor}}$$

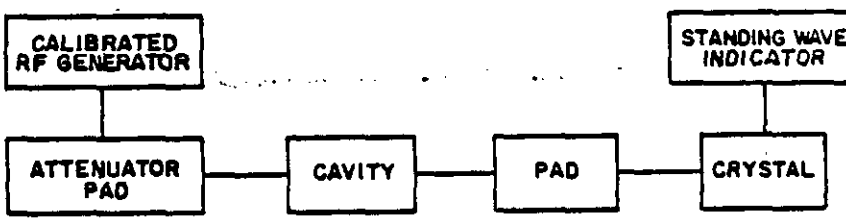
Where "n" is the water flow in gpm; "to" is the outlet water temperature and "ti" is the inlet water temperature in ^{o}C.

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NOTES: (Cont'd)

14. Input and Output Resonant Frequency

The RF frequency generator shall be adjusted to obtain a resonant peak on the standing wave indicator, and frequency shall be measured as in the block diagram shown below. (Be sure tube is making good contact with cavity.)



Input:

Two input resonant cavities, each mounted at the designated location of the tube, shall be in accordance with Figure 6.

Output:

Two identical output resonant cavities, each mounted at opposite ends of the tube, shall be in accordance with Figure 7.

15. Holding Period

Test to be performed at the conclusion of the holding period.

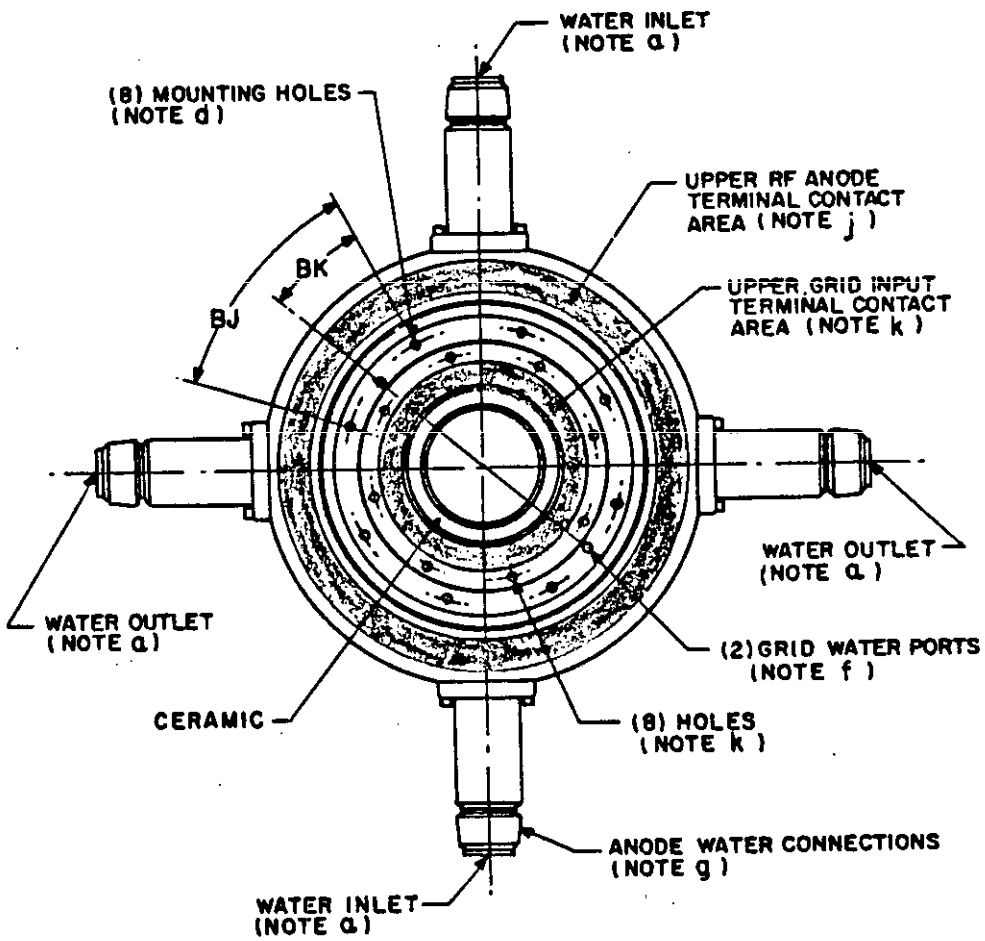
Custodian:
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Review activities:
Air Force - 11, 17
DSA - ES

Preparing activity:
Air Force - 80

(Project 5960-F326-6)

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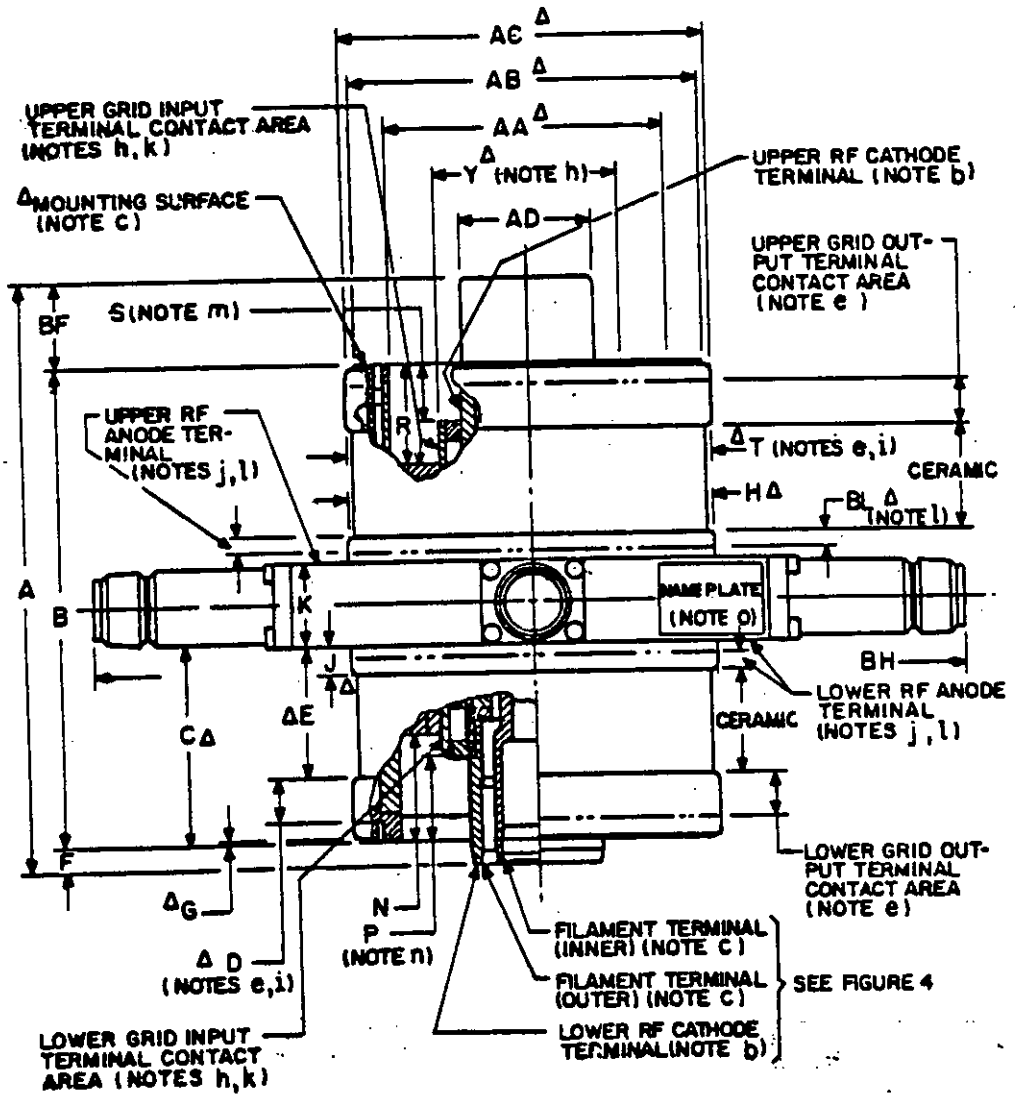


DIMENSIONS APPLY TO BOTH ENDS OF TUBE

END VIEW
UPPER-RF CATHODE-TERMINAL

FIGURE 1. Outline drawing Electron Tube type 7835

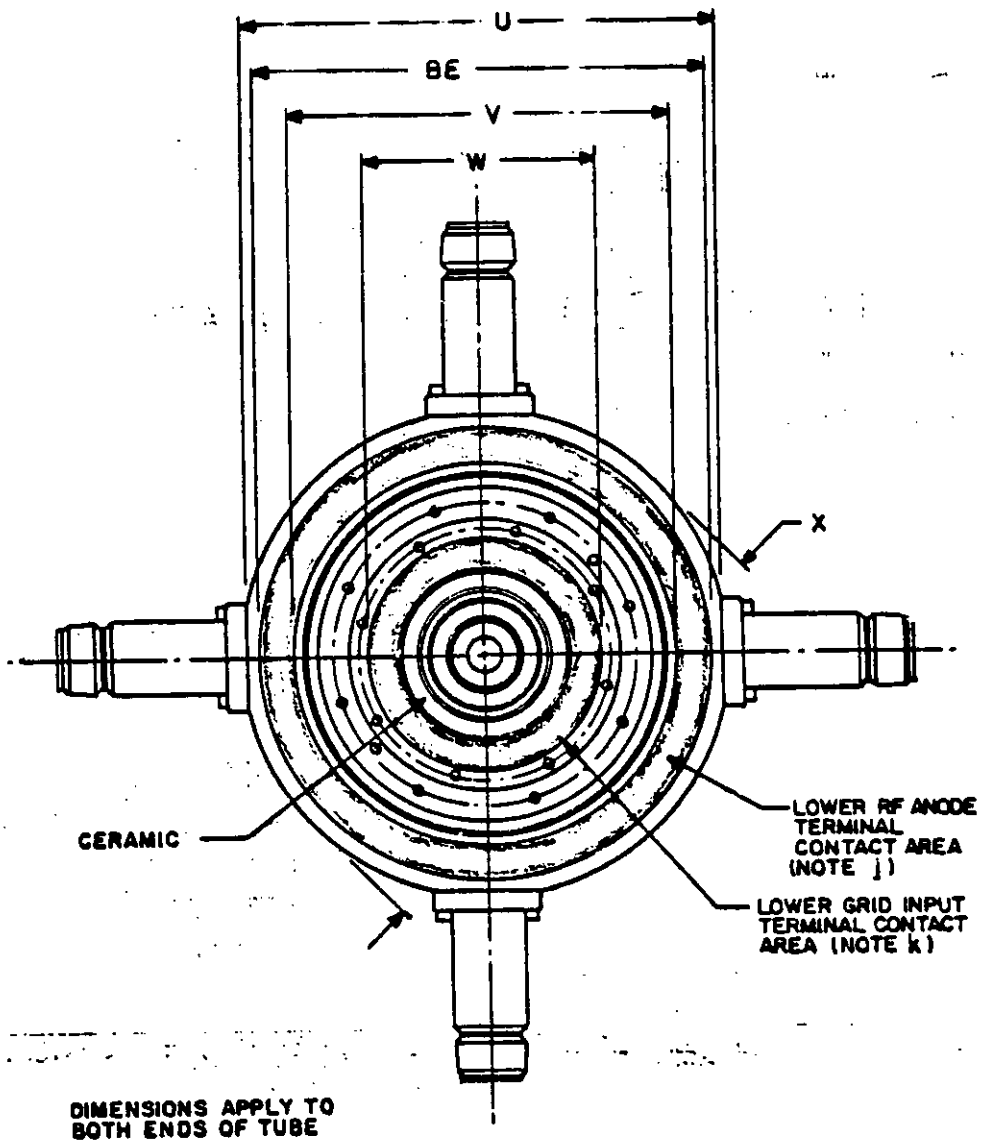
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Δ DIMENSIONS APPLY TO BOTH ENDS OF TUBE.

FIGURE 2. Outline drawing Electron Tube type 7835

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END VIEW
FILAMENT TERMINAL

FIGURE 3. Outline drawing Electron Tube type 7835

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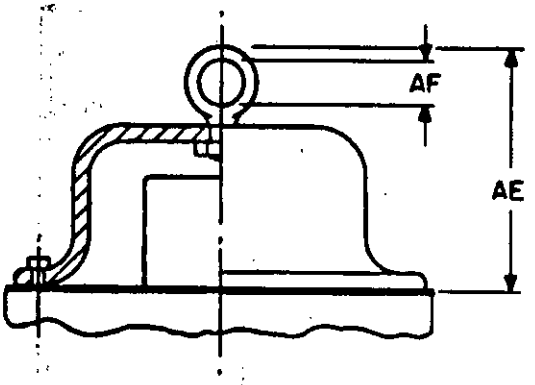


FIGURE 5 Detail of Lifting Plate (NOTE d)

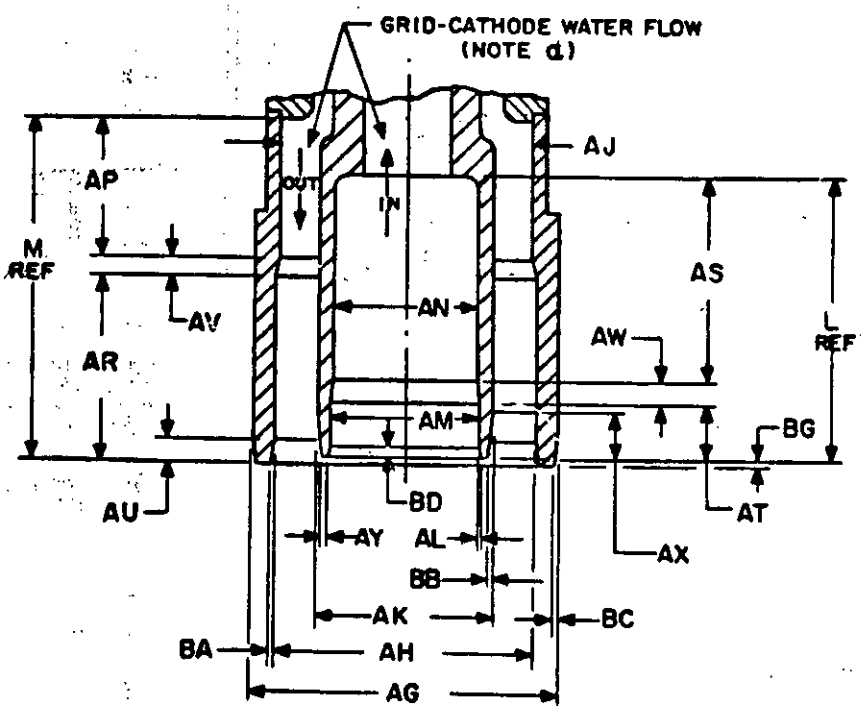


FIGURE 4. Detail of Filament Terminals

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DIMENSIONS

LTR	INCHES MIN	INCHES MAX	MILLIMETERS MIN	MILLIMETERS MAX	ANGLE DEGREES	NOTES
PREPRODUCTION						
A	-	17.00	-	431.8		
B	-	13.50	-	342.8		
D	.75	-	19.1	-		e,i
F	.52	.76	13.2	19.3		
G	-	.25	-	6.3		
H DIA	-	10.30	-	261.6		
J	.85	.96	21.6	24.4		
K	2.07	2.15	77.98	80.01		
L	3.34	3.41	84.84	86.87		
M	3.97	4.03	100.56	102.63		
N	2.92	3.03	74.2	76.9		
P	.50	-	12.7	-		h,n
R	2.52	2.63	64.0	66.8		
S	.93	-	23.62	-		h,m
U	13.55	13.80	344.2	350.5		
V DIA	-	12.00	-	304.8		j
W DIA	6.75	-	171.5	-		k
X DIA	13.85	14.10	351.8	358.1		
Y DIA	-	5.02	-	127.5		h
AA DIA	7.63	7.87	193.8	199.9		
AB DIA	9.40	9.64	238.8	244.8		
AE	-	8.00	-	152.4		
AF DIA	.875	-	22.23	-		
AK DIA	1.98	2.02	50.29	51.30		
AP	1.64	1.66	41.66	42.16		
AR	2.090	2.110	53.09	53.59		
AS	2.490	2.510	63.25	63.75		
AT	.615	.636	15.62	16.13		
BE DIA	13.40	-	340.4	-		j
BF	2.33	2.57	59.2	65.2		
BG	-	.10	-	2.5		
BH	-	24.00	-	609.6		
BK	-	-	-		22.5	
BL	.63	-	16.00	-		l

LTR	INCHES MIN	INCHES MAX	MILLIMETERS MIN	MILLIMETERS MAX	NOTES
QUALITY CONFORMANCE INSPECT., PART I					
C	5.51	5.68	140.0	144.3	
E	3.79	3.96	96.3	100.5	
T DIA	-	10.30	-	261.6	e,i
AC DIA	10.12	-	257.1	-	e
AD DIA	3.59	3.65	91.19	92.71	
AG DIA	3.59	3.65	91.19	92.71	
AH DIA	3.121	3.131	79.28	79.52	
AJ DIA	2.997	3.007	76.13	76.37	
AM DIA	1.745	1.755	44.33	44.57	
AN DIA	1.695	1.705	43.05	43.30	

REFERENCE DIMENSIONS

LTR	INCHES	MILLIMETERS	DEGREES	LTR
AL	.03	.76		
AU	.28	7.11		
AV	.25	6.35		
AW	.25	6.35		
AX	.50	12.7		
AY	.06	1.52		
BA	.04	1.02		
BB	.03	.76		
BC	.04	1.02		
BD	.10	2.5		
BJ	-	-	45	

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NOTES:

a. The anode and the grid-cathode water flows in this tube are directional as noted. Failure to observe correct flow direction may result in inadequate cooling and short tube life.

b. DC filament current must not be permitted to flow in the upper RF-cathode terminal. The internal structure of the tube is such that the potential of the upper RF-cathode terminal differs from that of the lower RF-cathode terminal by the amount of the DC filament voltage. The circuit designer should take care, therefore to avoid an external DC path between these two cathode terminals.

c. The tube should be operated vertically with either end up. The entire weight of the tube should be supported by the rf anode terminal. Care should be taken to avoid distortion or damage to the filament terminals by bumping or improperly fitting connectors. Total indicator run-out between terminals will not exceed 0.10" (2.54 mm).

d. The tube may be conveniently handled and moved by means of the lifting plate which may be attached to either mounting surface. The lifting plate should be removed prior to operating the tube. The mounting surface holes are .250"-20 UNC-1B x 0.25" (6.35 mm) minimum depth, equally spaced on a bolt circle of 8.75" (222.25 mm) diameter.

e. Along the tapered contact length D, dimension AC will increase from a minimum diameter of 10.15" (257.8 mm) (average diameter at this point is 10.20" (259.1 mm) to a maximum diameter T 10.30" (261.6 mm) (average diameter at T is 10.25" (260.4 mm). The maximum diameter T is at the end toward the ceramic.

f. The direction of flow, IN or OUT, is stamped at the side of each port. The ports are 0.250" $\pm$ 0.010" (6.35 $\pm$ 25 mm) diameter located 180° $\pm$ 1/2° apart on a circle of 8.25" (209.55 mm) diameter. The upper grid water ports are located on the mounting surface in the quadrant counter-clockwise from each anode water inlet, as viewed from the upper RF-cathode terminal end. The lower grid water ports are located on the mounting surface in the same quadrant, as viewed from upper RF-cathode terminal end.

g. The anode water connectors located 90° apart, as specially made and are similar in size to Hansen Plugs No. 12-T-46. Fittings may be obtained from the Hansen Mfg. Co., 4031 West 150th Street, Cleveland 11, Ohio.

h. Along the length S and P, the contact areas are tapered, increasing diameter 'Y' from a 4.90" (124.46 mm) average (4.88" (123.95 mm) minimum) at the recessed end to a 4.98" (125.98 mm) average (5.02" (127.5 mm) maximum) at the other end.

i. Circuit contacts should be made only over maximum length D 0.75" (19.1 mm) of the designated upper and lower output terminal contact areas as measured nearest the ceramic.

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NOTES: Cont'd

j. Contacts of the upper and lower RF anode terminal contact areas should be made at a diameter [illegible] than 13.40" (340.4 mm) or greater than 12.00" (304.8 mm).

k. Contact of the upper and lower grid input terminal contact areas should be made at a diameter less than 6.75" (171.45 mm). The holes located outside the contact area are 250"-20 UNC-1Bx 0.25" (6.35 mm) minimum depth, equally spaced on a bolt circle of 7.25" (184.15 mm) diameter.

l. Circuit contact should be made only over a maximum length, BL 0.63" (16.0 mm) of designated upper and lower RF-anode terminals as measured nearest the ceramic.

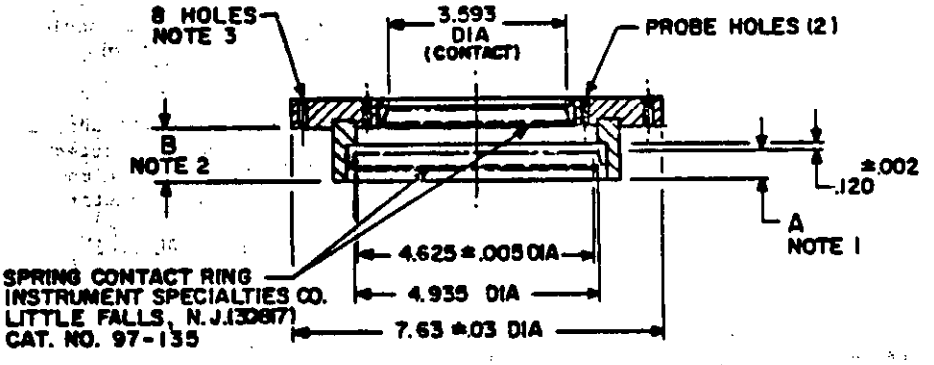
m. Circuit contact should be made only over maximum length S 0.93[illegible] (23.62 mm) of the designated upper grid input terminal.

n. Circuit contact should be made only over maximum length, P 0.50" (12.7 mm) of the designated lower grid input terminal.

o. Name plate may be located between any two adjacent water connections.

p. Metric equivalents are given for general information only and are based on 1 inch = 25.4 mm.

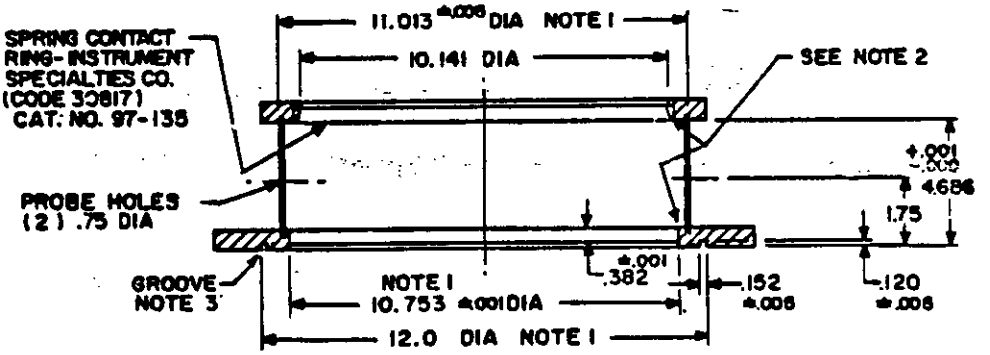
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NOTE -

1. DIMENSION A IS .61 ±.02 FOR CAVITY USED AT FILAMENT - TERMINALS END; 1.03 ± .20 FOR CAVITY USED AT UPPER - RF - CATHODE - TERMINAL END.
2. DIMENSION B IS 1.13 ± .20 FOR CAVITY USED AT FILAMENT - TERMINALS END; 1.55 ± .02 FOR CAVITY USED AT UPPER - RF - CATHODE - TERMINAL END
3. THESE HOLES PROVIDE THE MEANS OF EXTRACTING CAVITY.

FIGURE 6 Full Section View of Input Cavity
(for Input Strap - Resonant Frequency Test)



NOTES.-

1. RUNOUT ON DIAMETERS INDICATED NOT TO EXCEED .002
2. JUNTURE MUST PROVIDE GOOD RF ELECTRICAL CONTACT AROUND ITS ENTIRE CIRCUMFERENCE, SUCH AS BY A BRAZE OR SOFT SOLDER CONNECTION. MAX BRAZE OR SOLDER FILLET NOT TO EXCEED .030 RADIUS
3. GROOVE FOR BRAID CONTACT RING. THE BRAIDED RING, SUCH AS CAT. NO. 10-118, ELECTRONICS DIVISION, METAL TEXTILE CORP. ROSELLE, N.J. (CODE 82805). MUST INSURE CONTINUOUS CONTACT.

FIGURE 7 Full Section View of Output Cavity
(for Output Strap - Resonant Frequency Test)

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